

Task 1: Write a program to count the number of Is in a word entered by user. The count of number is in BCD. Convert the number into decimal number system and display the count on screen.

Code:

```

TITLE LAB 9 ACTIVITY 1                                ; Title of the program

.MODEL SMALL                                           ; Define the memory model as small
.STACK 100H                                           ; Set the stack size to 100H

.DATA
INPUT DB 'ENTER FOUR DIGIT NUMBER $'                ; Message prompt for the user
DIG1 DB ?                                             ; Variable to store the first digit
DIG2 DB ?                                             ; Variable to store the second digit
DIG3 DB ?                                             ; Variable to store the third digit
DIG4 DB ?                                             ; Variable to store the fourth digit

.CODE
MAIN PROC                                             ; Main procedure starts
    MOV AX, @DATA                                     ; Load the address of the data segment into AX
    MOV DS, AX                                       ; Set DS to the address of the data segment

    MOV BX, 0H                                       ; Initialize BX to zero

    MOV DX, OFFSET INPUT                             ; Load the address of the INPUT message into DX
    MOV AH, 9                                        ; Set the AH register for DOS display string function
    INT 21H                                          ; Call DOS interrupt to display the message

    MOV AH, 1                                        ; Set the AH register for DOS character input function
    INT 21H                                          ; Call DOS interrupt to input the first digit
    MOV DIG1, AL                                     ; Store the first digit

    MOV AH, 1                                        ; Set the AH register for DOS character input function
    INT 21H                                          ; Call DOS interrupt to input the second digit
    MOV DIG2, AL                                     ; Store the second digit

    MOV AH, 1                                        ; Set the AH register for DOS character input function
    INT 21H                                          ; Call DOS interrupt to input the third digit
    MOV DIG3, AL                                     ; Store the third digit

    MOV AH, 1                                        ; Set the AH register for DOS character input function
    INT 21H                                          ; Call DOS interrupt to input the fourth digit
    MOV DIG4, AL                                     ; Store the fourth digit

    MOV AL, DIG1                                     ; Move the first digit to AL
    MOV CX, 4                                        ; Set the loop counter to 4

SBACK1:
    ROR AL, 1                                        ; Rotate AL right through the carry flag
    JC COUNT1                                       ; Jump to COUNT1 if the carry flag is set
LBACK1:

```

```
LOOP SBACK1                                ; Repeat the loop

JMP NEXT1                                  ; Jump to NEXT1

COUNT1:
INC BX                                     ; Increment BX
JMP LBACK1                                ; Jump to LBACK1

NEXT1:
MOV AL, DIG2                              ; Move the second digit to AL
MOV CX, 4                                  ; Set the loop counter to 4

SBACK2:
ROR AL, 1                                 ; Rotate AL right through the carry flag
JC COUNT2                                 ; Jump to COUNT2 if the carry flag is set
LBACK2:                                    ; Label LBACK2
LOOP SBACK2                               ; Repeat the loop

JMP NEXT2                                  ; Jump to NEXT2

COUNT2:
INC BX                                     ; Increment BX
JMP LBACK2                                ; Jump to LBACK2

NEXT2:
MOV AL, DIG3                              ; Move the third digit to AL
MOV CX, 4                                  ; Set the loop counter to 4

SBACK3:
ROR AL, 1                                 ; Rotate AL right through the carry flag
JC COUNT3                                 ; Jump to COUNT3 if the carry flag is set
LBACK3:                                    ; Label LBACK3
LOOP SBACK3                               ; Repeat the loop

JMP NEXT3                                  ; Jump to NEXT3

COUNT3:
INC BX                                     ; Increment BX
JMP LBACK3                                ; Jump to LBACK3

NEXT3:
MOV AL, DIG4                              ; Move the fourth digit to AL
MOV CX, 4                                  ; Set the loop counter to 4

SBACK4:
ROR AL, 1                                 ; Rotate AL right through the carry flag
JC COUNT4                                 ; Jump to COUNT4 if the carry flag is set
LBACK4:                                    ; Label LBACK4
LOOP SBACK4                               ; Repeat the loop

JMP GO                                     ; Jump to GO

COUNT4:
```

```
INC BX      ; Increment BX
JMP LBACK4  ; Jump to LBACK4
```

GO:

```
MOV AL, BL      ; Move the result (number of set bits) to AL
MOV DL, 01H     ; Move 01H to DL
MUL DL          ; Multiply AL by DL
AAM             ; Convert AL to ASCII
OR AX, 3030H    ; Convert the result to ASCII
MOV BX, AX      ; Move the ASCII result to BX

MOV AH, 2       ; Set the AH register for DOS display character function
MOV DL, 0DH     ; Load the carriage return character into DL
INT 21H         ; Call DOS interrupt to display the carriage return

MOV DL, 0AH     ; Load the linefeed character into DL
INT 21H         ; Call DOS interrupt to display the linefeed

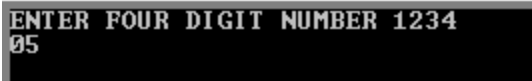
MOV AH, 2       ; Set the AH register for DOS display character function
MOV DL, BH      ; Move the tens place (BH) to DL
INT 21H         ; Call DOS interrupt to display the tens place

MOV AH, 2       ; Set the AH register for DOS display character function
MOV DL, BL      ; Move the ones place (BL) to DL
INT 21H         ; Call DOS interrupt to display the ones place

MOV AH, 4CH     ; Set the AH register for DOS exit function
INT 21H         ; Call DOS interrupt to exit the program

MAIN ENDP      ; End of main procedure
END MAIN       ; End of the program
```

Output:

 emulator screen (80x25 chars)

ENTER FOUR DIGIT NUMBER 1234
05

In BCD codes:

1 = 0001,
2 = 0010,
3 = 0011,
4 = 0100,

Total number of 1s = 05

Task 2: Take two 1-digit inputs (from 0 to 9) from user through keyboard. Write a program to perform the following operations, and display the results for all the operations on screen.

- Addition
- Subtraction
- Multiplication
- Division

• **Addition**

Code:

```

TITLE LAB 9 ACTIVITY 2A                ; Title of the program
.MODEL SMALL                          ; Define the memory model as small
.STACK 100H                           ; Set the stack size to 100H

.DATA
FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ; Message to prompt user for the first number
SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ; Message to prompt user for the second
number
RESULT DB 0DH, 0AH, 'THE RESULT IS = $' ; Message to display the result

.CODE                                ; Beginning of the code segment

MAIN PROC                            ; Start of the main procedure
    MOV AX,@DATA                    ; Move the address of data segment to AX
    MOV DS,AX                      ; Initialize DS register with the data segment address

    LEA DX,FIRST_NUMBER            ; Load effective address of the first number prompt to DX
    MOV AH,9                      ; DOS function to display a string
    INT 21H                       ; Call DOS interrupt

    MOV AH,1                      ; DOS function to read a character from standard input
    INT 21H                       ; Call DOS interrupt
    MOV BL,AL                     ; Move the input to BL

    LEA DX,SECOND_NUMBER           ; Load effective address of the second number prompt to DX
    MOV AH,9                      ; DOS function to display a string
    INT 21H                       ; Call DOS interrupt

    MOV AH,1                      ; DOS function to read a character from standard input
    INT 21H                       ; Call DOS interrupt

    MOV AH,00H                    ; Clear AH register
    XCHG AL,BL                    ; Exchange AL and BL
    ADD AL,BL                     ; Add BL to AL
    AAA                          ; ASCII adjust AL after addition

    OR AX,3030H                  ; Convert result to ASCII in AX
    MOV BX,AX                    ; Move the ASCII value to BX

    LEA DX,RESULT                 ; Load effective address of the result message to DX
    MOV AH,9                     ; DOS function to display a string

```

```

INT 21H                ; Call DOS interrupt

MOV AH,2               ; DOS function to display a character
MOV DL,BH              ; Move the high digit of the result to DL
INT 21H                ; Call DOS interrupt

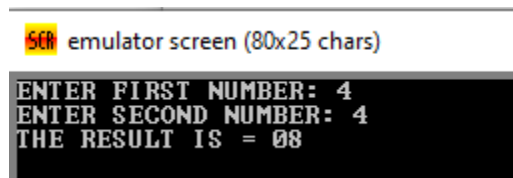
MOV AH,2               ; DOS function to display a character
MOV DL,BL              ; Move the low digit of the result to DL
INT 21H                ; Call DOS interrupt

MOV AH,4CH             ; DOS function to exit the program
INT 21H                ; Call DOS interrupt

MAIN ENDP              ; End of the main procedure
END MAIN                ; End of the program

```

Output:



```

emulator screen (80x25 chars)
ENTER FIRST NUMBER: 4
ENTER SECOND NUMBER: 4
THE RESULT IS = 08

```

• Subtraction

Code:

```

TITLE LAB 9 ACTIVITY 2B          ; Title of the program
.MODEL SMALL                     ; Define the memory model as small
.STACK 100H                      ; Set the stack size to 100H

.DATA
FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ; Message to prompt user for the first number
SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ; Message to prompt user for the second
number
RESULT DB 0DH, 0AH, 'THE RESULT IS = $' ; Message to display the result

.CODE                             ; Beginning of the code segment

MAIN PROC                         ; Start of the main procedure
    MOV AX,@DATA                  ; Move the address of data segment to AX
    MOV DS,AX                     ; Initialize DS register with the data segment address

    LEA DX,FIRST_NUMBER           ; Load effective address of the first number prompt to DX
    MOV AH,9                      ; DOS function to display a string
    INT 21H                       ; Call DOS interrupt

    MOV AH,1                      ; DOS function to read a character from standard input
    INT 21H                       ; Call DOS interrupt
    MOV BL,AL                     ; Move the input to BL

    LEA DX,SECOND_NUMBER          ; Load effective address of the second number prompt to DX

```

```

MOV AH,9                ; DOS function to display a string
INT 21H                 ; Call DOS interrupt

MOV AH,1                ; DOS function to read a character from standard input
INT 21H                 ; Call DOS interrupt

MOV AH,00H              ; Clear AH register
XCHG AL,BL              ; Exchange AL and BL
SUB AL,BL               ; Subtract BL from AL
AAS                     ; ASCII adjust AL after subtraction

OR AX,3030H             ; Convert result to ASCII in AX
MOV BX,AX               ; Move the ASCII value to BX

LEA DX,RESULT           ; Load effective address of the result message to DX
MOV AH,9                ; DOS function to display a string
INT 21H                 ; Call DOS interrupt

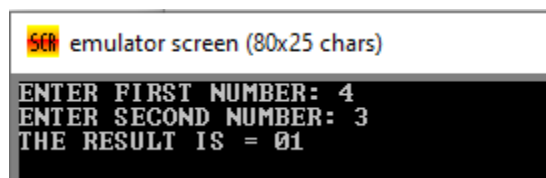
MOV AH,2                ; DOS function to display a character
MOV DL,BH               ; Move the high digit of the result to DL
INT 21H                 ; Call DOS interrupt

MOV AH,2                ; DOS function to display a character
MOV DL,BL               ; Move the low digit of the result to DL
INT 21H                 ; Call DOS interrupt

MOV AH,4CH              ; DOS function to exit the program
INT 21H                 ; Call DOS interrupt

MAIN ENDP               ; End of the main procedure
END MAIN                ; End of the program

```

Output:


```

emulator screen (80x25 chars)
ENTER FIRST NUMBER: 4
ENTER SECOND NUMBER: 3
THE RESULT IS = 01

```

- **Multiplication**

Code:

```

TITLE LAB 9 ACTIVITY 2C      ; Title of the program
.MODEL SMALL                 ; Define the memory model as small
.STACK 100H                  ; Set the stack size to 100H

.DATA
FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ; Message to prompt user for the first number
SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ; Message to prompt user for the second
number
RESULT DB 0DH, 0AH, 'THE RESULT IS = $' ; Message to display the result

```

```
.CODE                                     ; Beginning of the code segment

MAIN PROC                               ; Start of the main procedure
    MOV AX,@DATA                         ; Move the address of data segment to AX
    MOV DS,AX                           ; Initialize DS register with the data segment address

    LEA DX,FIRST_NUMBER                 ; Load effective address of the first number prompt to DX
    MOV AH,9                            ; DOS function to display a string
    INT 21H                             ; Call DOS interrupt

    MOV AH,1                            ; DOS function to read a character from standard input
    INT 21H                             ; Call DOS interrupt
    AND AL,0FH                          ; Mask high bits to get the numeric value
    MOV BL,AL                           ; Move the input to BL

    LEA DX,SECOND_NUMBER                 ; Load effective address of the second number prompt to DX
    MOV AH,9                            ; DOS function to display a string
    INT 21H                             ; Call DOS interrupt

    MOV AH,1                            ; DOS function to read a character from standard input
    INT 21H                             ; Call DOS interrupt

    MOV AH,00H                          ; Clear AH register
    AND AL,0FH                          ; Mask high bits to get the numeric value
    MUL BL                              ; Multiply AX by BL
    AAM                                 ; ASCII adjust AX to unpack binary-coded decimal

    OR AX,3030H                         ; Convert result to ASCII in AX
    MOV BX,AX                           ; Move the ASCII value to BX

    LEA DX,RESULT                        ; Load effective address of the result message to DX
    MOV AH,9                            ; DOS function to display a string
    INT 21H                             ; Call DOS interrupt

    MOV AH,2                            ; DOS function to display a character
    MOV DL,BH                           ; Move the high digit of the result to DL
    INT 21H                             ; Call DOS interrupt

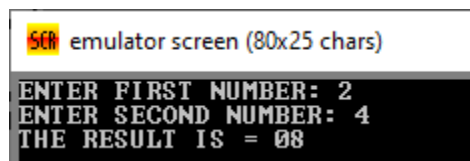
    MOV AH,2                            ; DOS function to display a character
    MOV DL,BL                           ; Move the low digit of the result to DL
    INT 21H                             ; Call DOS interrupt

    MOV AH,4CH                          ; DOS function to exit the program
    INT 21H                             ; Call DOS interrupt

MAIN ENDP                               ; End of the main procedure
```

END MAIN ; End of the program

Output:



```

SCR emulator screen (80x25 chars)
ENTER FIRST NUMBER: 2
ENTER SECOND NUMBER: 4
THE RESULT IS = 08
  
```

• Division

Code:

```

TITLE LAB 9 ACTIVITY 2D                ; Title of the program
.MODEL SMALL                          ; Define the memory model as small
.STACK 100H                           ; Set the stack size to 100H

.DATA
FIRST_NUMBER DB 'ENTER FIRST NUMBER : $' ; Message to prompt user for the first number
SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER : $' ; Message to prompt user for the second
number
STR1 DB 0DH, 0AH, 'REMAINDER IS = $' ; Message to display the remainder
STR2 DB 0DH, 0AH, 'QUOTIENT IS = $' ; Message to display the quotient

.CODE                                ; Beginning of the code segment

MAIN PROC                            ; Start of the main procedure
    MOV AX,@DATA                    ; Move the address of data segment to AX
    MOV DS,AX                      ; Initialize DS register with the data segment address

    LEA DX,FIRST_NUMBER            ; Load effective address of the first number prompt to DX
    MOV AH,9                      ; DOS function to display a string
    INT 21H                        ; Call DOS interrupt

    MOV AH,1                      ; DOS function to read a character from standard input
    INT 21H                        ; Call DOS interrupt
    AND AL,0FH                    ; Mask high bits to get the numeric value
    MOV BL,AL                     ; Move the input to BL

    LEA DX,SECOND_NUMBER          ; Load effective address of the second number prompt to DX
    MOV AH,9                      ; DOS function to display a string
    INT 21H                        ; Call DOS interrupt

    MOV AH,1                      ; DOS function to read a character from standard input
    INT 21H                        ; Call DOS interrupt

    MOV AH,00H                    ; Clear AH register
    AND AL,0FH                    ; Mask high bits to get the numeric value
    XCHG AL,BL                    ; Exchange AL and BL registers
    AAD                           ; ASCII adjust AX to unpack binary-coded decimal
  
```

```

DIV BL          ; Divide AX by BL; quotient in AL, remainder in AH
OR AX,3030H     ; Convert quotient to ASCII in AX
MOV BX,AX       ; Move the ASCII value to BX

LEA DX,STR2     ; Load effective address of the quotient message to DX
MOV AH,9        ; DOS function to display a string
INT 21H         ; Call DOS interrupt

MOV AH,2        ; DOS function to display a character
MOV DL,BL       ; Move the quotient to DL
INT 21H         ; Call DOS interrupt

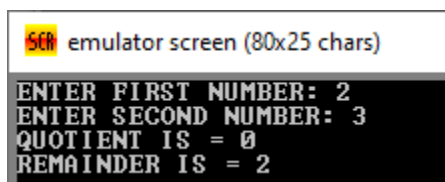
LEA DX,STR1     ; Load effective address of the remainder message to DX
MOV AH,9        ; DOS function to display a string
INT 21H         ; Call DOS interrupt

MOV AH,2        ; DOS function to display a character
MOV DL,BH       ; Move the remainder to DL
INT 21H         ; Call DOS interrupt

MOV AH,4CH      ; DOS function to exit the program
INT 21H         ; Call DOS interrupt

MAIN ENDP      ; End of the main procedure
END MAIN       ; End of the program

```

Output:


```

emulator screen (80x25 chars)
ENTER FIRST NUMBER: 2
ENTER SECOND NUMBER: 3
QUOTIENT IS = 0
REMAINDER IS = 2

```

Task 3: Write a program to make a counter that counts up from 0 to 99 in BCD and display the result line by line.

Code:

```

TITLE LAB 9 ACTIVITY 3          ; Title of the program

.MODEL SMALL                   ; Define the memory model as small
.STACK 100H                    ; Set the stack size to 100H

.CODE
MAIN PROC                      ; Main procedure starts
    MOV BL, 0                  ; Initialize BL register to 0

```

AGAIN:

```

MOV DL, 1           ; Load 1 into DL register
MOV AL, BL          ; Move the value in BL to AL register

MUL DL              ; Multiply AL by DL, result in AX
AAM                 ; ASCII adjust AX to separate digits

OR AX, 3030H        ; Convert the digits to ASCII
MOV CX, AX          ; Move the ASCII digits to CX register

MOV AH, 2           ; Set the AH register for DOS display character function
MOV DL, CH          ; Move the high-order digit to DL
INT 21H             ; Call DOS interrupt to display the digit

MOV DL, CL          ; Move the low-order digit to DL
INT 21H             ; Call DOS interrupt to display the digit

MOV AH, 2           ; Set the AH register for DOS display character function
MOV DL, 0AH         ; Move the linefeed character to DL
INT 21H             ; Call DOS interrupt to display the linefeed

MOV DL, 0DH         ; Move the carriage return character to DL
INT 21H             ; Call DOS interrupt to display the carriage return

INC BL              ; Increment BL
CMP BL, 100         ; Compare BL with 100
JNE AGAIN           ; Jump to AGAIN if not equal

MOV AH, 4CH         ; Set the AH register for DOS exit function
INT 21H             ; Call DOS interrupt to exit the program

MAIN ENDP           ; End of main procedure
END MAIN            ; End of the program

```

Output:

Scr emulator screen (80x25 chars)

```

00
01
02
03
04
05
06
07
08
09
10
11
12
13
14
15
16
17
18
19
20
21
22
23

```

clear screen

change font

0/16

Scr emulator screen (80x25 chars)

```

77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99

```

clear screen

change font

0/16

Task 4: Write a program to ask user to enter his/her name in lower-case, convert them in upper-case, and display the name on screen. Use buffered input service "0AH" of "INT 21H" to take input from user.

Code:

```

TITLE LAB 9 ACTIVITY 4                ; Title of the program

.MODEL SMALL                          ; Define the memory model as small
.STACK 100H                           ; Set the stack size to 100H

.DATA
FIRSTNAME DB 'ENTER FIRST NAME IN LOWER CASE : $'    ; Message prompt for the user
INPUTNAME DB 10, ?, 10 DUP (?)                      ; Buffer to store user input
UPPERCASE DB 8 DUP ('$')                            ; Buffer to store uppercase version
DISPLAY DB 'YOUR FIRST NAME IS: $'                  ; Message to display the result

.CODE
MAIN PROC                               ; Main procedure starts
    MOV AX, @DATA                        ; Load the address of the data segment into AX
    MOV DS, AX                          ; Set DS to the address of the data segment

    MOV AH, 9                            ; Set the AH register for DOS display string function
    LEA DX, FIRSTNAME                    ; Load the address of the FIRSTNAME message into DX
    INT 21H                             ; Call DOS interrupt to display the message

    MOV AH, 0AH                          ; Set the AH register for DOS buffered input function
    MOV DX, OFFSET INPUTNAME             ; Load the address of INPUTNAME buffer into DX
    INT 21H                             ; Call DOS interrupt to input the name

    MOV SI, OFFSET INPUTNAME              ; Load the address of INPUTNAME into SI
    MOV BX, OFFSET UPPERCASE              ; Load the address of UPPERCASE into BX

    MOV CX, 7                            ; Set the loop counter to 7 (assuming 8-character name)

AGAIN:
    MOV AL, [SI] + 2                      ; Load the next character from INPUTNAME into AL
    CMP AL, 61H                          ; Compare AL with ASCII 'a'
    JB OVER                              ; Jump to OVER if AL is less than 'a'
    CMP AL, 7AH                          ; Compare AL with ASCII 'z'
    JA OVER                              ; Jump to OVER if AL is greater than 'z'
    AND AL, 11011111B                    ; Convert lowercase to uppercase by clearing the 6th bit
    OVER:
    MOV [BX], AL                          ; Store the modified character in UPPERCASE
    INC SI                                ; Increment SI to point to the next character
    INC BX                                ; Increment BX to point to the next position in UPPERCASE
    LOOP AGAIN                            ; Repeat the loop

    MOV AH, 2                            ; Set the AH register for DOS display character function
    MOV DL, 0AH                          ; Load the linefeed character into DL

```

INT 21H	; Call DOS interrupt to display the linefeed
MOV DL, 0DH	; Load the carriage return character into DL
INT 21H	; Call DOS interrupt to display the carriage return
MOV AH, 9	; Set the AH register for DOS display string function
LEA DX, DISPLAY	; Load the address of the DISPLAY message into DX
INT 21H	; Call DOS interrupt to display the message
MOV AH, 9	; Set the AH register for DOS display string function
MOV DX, OFFSET UPPERCASE	; Load the address of UPPERCASE into DX
INT 21H	; Call DOS interrupt to display the uppercase name
MOV AH, 4CH	; Set the AH register for DOS exit function
INT 21H	; Call DOS interrupt to exit the program
MAIN ENDP	; End of main procedure
END MAIN	; End of the program

Output:

 emulator screen (80x25 chars)

```
ENTER FIRST NAME IN LOWER CASE: faisal
YOUR FIRST NAME IS: FAISAL
```